

PART A
(25 marks)

1. Evaluate the following limits (if exists).

(a) $\lim_{x \rightarrow 2} \frac{6(x-2)}{x^3-8}$ (3 marks)

(b) $\lim_{x \rightarrow 1} \frac{e^{2\left(\frac{x-1}{x+1}\right)} - 1}{e^{\left(\frac{x-1}{x+1}\right)} - 1}$ (3 marks)

(c) $\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2+1}}{3x-1}$ (3 marks)

2. (a) Find the derivative of $f(x) = \frac{1}{2x-3}$ with respect to x , using the first principle.

(4 marks)

- (b) Differentiate each of the following function with respect to x .

i) $f(x) = \ln\left(\frac{x+1}{x^2+2x}\right)$ (2 marks)

ii) $f(x) = 3^{2x-1} + e^{3x}$ (2 marks)

iii) $f(x) = 4 \tan x + \frac{1}{x} + 3x$ (3 marks)

3. The equation of a curve is $y = \frac{1}{x} - 4x^2$. Find the coordinates of the stationary points and determine its nature.

(5 marks)

PART B

(75 marks)

1. (a) Given that $a = \log_x 2$, $b = \log_x 5$. Express $\log_x \left(\frac{4}{25\sqrt{x}} \right)$ in terms of a and b . (4 marks)
- (b) If $3i(2i^3 + i^2) = a + bi$ where a, b are real numbers, determine the values of a and b . Hence, express $a + bi$ in polar form. (6 marks)
2. (a) Solve $\log(2x + 3) - 2\log 2 = 2 + \log\left(\frac{x}{20}\right)$ (11 marks)
- (b) Solve $\left| \frac{x+3}{x} \right| > 2$
3. Given $f(x) = 2e^x + 1$
- (a) Show that $f(x)$ is a one to one function. (2 marks)
- (b) Find $f^{-1}(x)$ (3 marks)
- Hence, sketch $f(x)$ and $f^{-1}(x)$ on the same axis and state the domain and range of both functions. (5 marks)
4. Function f and g are defined by
- $$f(x) = \sqrt{2-x}, \quad x < 2$$
- $$g(x) = 2 - x^2, \quad x > 0$$
- (a) Find the composite function $f \circ g$ and $g \circ f$ (6 marks)
- (b) State the domain of $f \circ g(x)$ (1 mark)
- (c) State the relationship between $f(x)$ and $g(x)$. (1 mark)
5. (a) State the conditions for continuity of $f(x)$ at $x = a$. (3 marks)

- (b) The function f is defined by

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x < 1 \\ m^2 - 2, & x = 1 \\ \frac{|1 - x|}{x - 1} + n, & x > 1 \end{cases}$$

where m and n are constants.

- (i) If $\lim_{x \rightarrow 1} f(x)$ exists, find the value of n . (6 marks)
- (ii) If f is continuous at $x = 1$, determine the values of m . (2 marks)
6. (a) The parametric equations of a curve are $x = \theta - \sin \theta$ and $y = 1 - \cos \theta$. Find the equation of the normal to the curve at a point with parameter $\frac{\pi}{2}$. (7 marks)
- (b) Find the value of $\frac{dy}{dx}$ at the point $(1, 0)$ which lies on the curve $xe^y + y \ln x = 5$. (9 marks)
7. A drop of oil is dropped on a stagnant water surface. The oil drop makes a circular mark which starts to increase in size. The radius r , in cm, of the circular mark is given by $r = \frac{4(1+t^2)}{9+t^2}$, where t in minutes, after the oil is dropped on the water surface.
- (a) Find the value of t when $r = \frac{4}{5}$. (2 marks)
- (b) Express $\frac{dr}{dt}$ in terms of t . (3 marks)
- (c) Calculate the rate of change of the area of the circular mark when $r = \frac{4}{5}$. (4 marks)

END OF QUESTIONS PAPER

Answers : PART A

1. (a) $\frac{1}{2}$ (b) 2 (c) -1

2. (a) $\frac{-2}{(2x-3)^2}$

(b) (i) $f'(x) = \frac{1}{x+1} - \frac{(2x+2)}{x^2+2x}$

(ii) $f'(x) = 3^{2x-1}(2) \ln 3 + 3e^{3x}$

(iii) $f'(x) = 4 \sec^2 x - \frac{1}{x^2} + 3$

3. $\left(-\frac{1}{2}, -3\right)$ is a maximum point

Answers : PART B

1. (a) $2a - 2b - \frac{1}{2}$ (b) $a = 6, b = -3, \therefore z = \sqrt{45}(\cos(-0.148\pi) + i \sin(-0.148\pi))$

2. $\{x: -1 < x < 0 \text{ or } 0 < x < 3\}$

3. (b) $f^{-1}(x) = \ln\left(\frac{x-1}{2}\right)$

$D_f : (-\infty, \infty) \quad R_f : (1, \infty)$

$D_{f^{-1}} : (1, \infty) \quad R_{f^{-1}} : (-\infty, \infty)$

4. (a) $f \circ g = x \quad g \circ f = x$ (b) $D_{f \circ g} : (0, 2)$ (c) f and g are inverse each other.

5. (a) (i) f(a) is defined (ii) $\lim_{x \rightarrow a} f(x)$ exists (iii) $\lim_{x \rightarrow a} f(x) = f(a)$

(b) (i) n = 1 (ii) m = -2, m = 2

6. (a) $y = -x + \frac{\pi}{2}$ (b) $\frac{dy}{dx} = -1$

7. (a) t = 1

(b) $\frac{dr}{dt} = \frac{64t}{(9+t^2)^2}$

(c) $\frac{dA}{dt} = \frac{128\pi}{125} \text{ cm}^2 \text{ s}^{-1}$